

Hypercapnia

Last updated February 19, 2026  19 min read  ScholarRx

Adapted by: Ian Murray, Ryan Downey, Patrick Brooks, Eleanor Schrems

Editors: Jordan Finn, Russell Johnson

 Adapted from: [Hypercapnia](#)

Listen to this Brick

0:00 / 32:16 1x

Learning Objectives (5)

Versions

After completing this brick, you will be able to:

- 1 Describe the clinical presentation of hypercapnia.
- 2 Explain the mechanisms underlying hypercapnia.
- 3 Explain the consequences of hypercapnia and the body's compensatory mechanisms.
- 4 Interpret clinical findings to properly diagnose hypercapnia.
- 5 Outline the management of hypercapnia.



CASE CONNECTION

NK is a 56-year-old male with chronic obstructive pulmonary disease (COPD) due to long-term heavy smoking. He uses supplemental oxygen at home. He now comes to the emergency department with worsening dyspnea. The initial evaluation suggests a COPD exacerbation. He is treated with bronchodilators and supplemental oxygen. With administered oxygen, his oxygen saturation increases from 89% to 95%. However, his estimated partial pressure of arterial carbon dioxide (PaCO_2) as measured by capnography rises from 45 mm Hg to 70 mm Hg. NK now appears confused.

What explains the change in his lab values and clinical condition after treatment? Consider your answer as you read, and we'll revisit NK at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Is Hypercapnia (LO1)?

Hypercapnia (sometimes called hypercarbia) is an elevation of the partial pressure of carbon dioxide in the arterial blood (PaCO_2). It is typically defined as a PaCO_2 level of more than 42 to 45 mmHg. This elevation is concerning because:

- CO₂ is the main stimulus for the respiratory drive in most people.
- Hypercapnia is often a marker for severe pulmonary disease (due to failed ventilation) or for central nervous system (CNS) disorders (due to depression of the respiratory centers in the brain), such as in opioid overdose.
- Hypercapnia often causes respiratory acidosis, which is a cause of acidemia (low pH) if the CO₂ elevation is sufficiently severe or acute. The acidemia can cause abnormal cardiac and nerve conduction.

Patients with severe hypercapnia may require intubation and ventilation, and the PaCO₂ is followed closely in patients in the intensive care unit (ICU).

How Do Patients With Hypercapnia Present (LO2)?

Patients with hypercapnia often have histories of pulmonary disease (eg, chronic obstructive pulmonary disease [COPD]) but may also have disorders that disrupt their CNS drive to breathe (eg, opioid or barbiturate overdoses, head trauma, or other CNS disease).

These patients are often quite short of breath (dyspneic). Sometimes this occurs because they also have hypoxemia as part of their lung disease. However, even without hypoxemia, an elevated PaCO_2 is an extremely noxious stimulus to the CNS respiratory centers, and even mild elevations will provoke a feeling of dyspnea. It will also trigger hyperventilation to correct the hypercapnia, by breathing off the excess CO_2 . Other symptoms may include flushing with palmar erythema (redness due to vasodilation) and encephalopathy with asterixis (flapping movements of hands, feet or head) or decreased mental state, even coma.

However, some patients may not have any symptoms. Unlike hypoxemia, the body can safely tolerate mild degrees of hypercapnia, and even severe hypercapnia if it is chronic.

drive to breathe in patients with hypercapnia.

conveys the sense of shortness of breath as part of the increased
Carbon dioxide is a potent central nervous system stimulus and

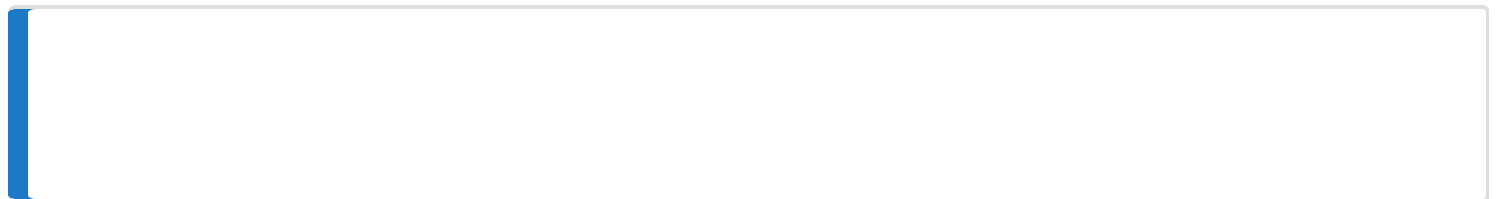
What Are the Main Causes of Hypercapnia (LO2)?

Most cases of hypercapnia are from **reduced ventilatory removal of CO₂ by the lungs**. Increased production of CO₂ (from, e.g., exercise, stress, and infection) can contribute to the PaCO₂ elevation but rarely causes hypercapnia on its own. The two main reasons for decreased CO₂ excretion by the lungs are:

- Reduced minute ventilation (most common).
- Ventilation/perfusion ($\dot{V}/\dot{Q}$) mismatching, including increased pathologic dead space.

Reduced Minute Ventilation

Some causes of hypercapnia are a decrease in the volume of air moving in and out of the respiratory system per unit time. The equation for minute ventilation is:



where V_E represents minute ventilation; RR, respiratory rate; and V_T , tidal volume.

This means that when either the respiratory rate or tidal volume is reduced, minute ventilation initially drops. Often, if one drops (e.g., the rate), a compensatory increase occurs in the other (e.g., the tidal volume increases). However, when this compensation fails, minute ventilation can be

persistently reduced. The partial pressure of alveolar carbon dioxide (PACO_2) then increases because it cannot be excreted during ventilation. Because the PACO_2 is equal to the PaCO_2 (CO_2 is very soluble and easily diffusible across the alveolar-capillary membrane), the PaCO_2 also rises, causing hypercapnia.

Decreased Respiratory Rate. This is mostly due to inhibition of the CNS respiratory centers. Causes include:

- Head trauma
- Drug intoxications (e.g., opioids, alcohol, other sedatives)
- Brain herniation or increased cerebral pressure

Note that most pulmonary diseases result in increased respiratory rate because they cause hypoxemia. However, sometimes, COPD, severe asthma, or pneumonia can present with reduced respiratory rate due to fatigue from the increased work of breathing.

Decreased Tidal Volume. Tidal volume can decrease because CNS depression decreases the strength of respiratory muscle contraction. However, the most common cause of reduced tidal volume is an upper or lower respiratory tract problem or disease:

- Airway obstruction due to trauma, edema in burns and anaphylaxis, airway secretions, bronchoconstriction in COPD and asthma, and excess neck mass in sleep apnea
- Respiratory muscle fatigue from increased work of breathing in acute COPD and asthma exacerbation, acute respiratory distress syndrome (ARDS), and increased metabolic demand in sepsis or critical illness
- Shallow breathing due to weakness of respiratory muscles in myasthenia gravis, botulism, Guillain-Barré syndrome, phrenic nerve palsy, or neuromuscular blocker toxicity

- Shallow breathing from chest wall deformities such as kyphoscoliosis, pectus excavatum, rib fracture, and ankylosing spondylitis, restricting chest wall expansion (however, these are mostly chronic disorders, so they are usually better compensated)

V/Q Mismatch

Effective gas exchange requires well-matched lung perfusion ($\dot{Q}$) and ventilation ($\dot{V}$). Decreases in $\dot{V}$ or $\dot{Q}$, leading to $\dot{V}/\dot{Q}$ mismatch, most often cause hypoxemia, but in some cases can also cause hypercapnia.

Underventilation of Multiple Lung Segments. The most common type of $\dot{V}/\dot{Q}$ mismatch causing hypercapnia is characterized by underventilation of multiple lung segments. An example, due to mucus plugging, is shown in Figure 1.

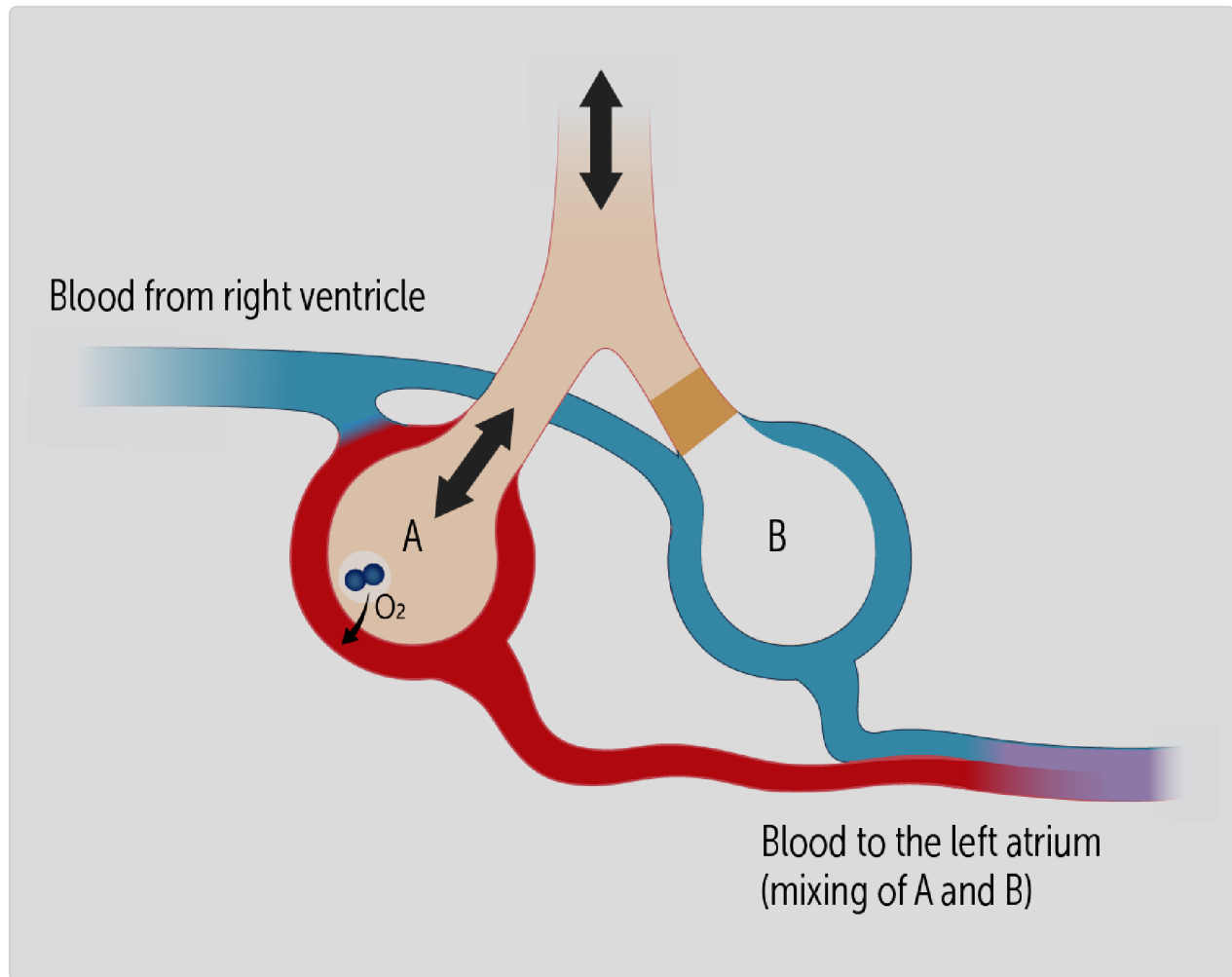


Figure 2 $\dot{V}/\dot{Q}$ mismatch in pulmonary disease

Hypercapnia due to $\dot{V}/\dot{Q}$ mismatch with reduced ventilation can occur in many pulmonary diseases, either by loss of alveoli (eg, in bronchitis/COPD) or by fluid entering the alveoli (e.g., in pneumonia, cardiogenic pulmonary edema, ARDS, cystic fibrosis).

Oxygen Therapy. Interestingly, hypercapnia due to $\dot{V}/\dot{Q}$ mismatch can also be caused by aggressive oxygen treatment of COPD. In COPD, pulmonary arterioles are normally constricted to try to shunt blood away from poorly ventilated areas of the lung (**hypoxic pulmonary vasoconstriction**), minimizing $\dot{V}/\dot{Q}$ mismatch. Oxygen therapy can reverse this compensation, creating mismatched areas of ventilation and perfusion (now with reduced perfusion to some areas) and causing reduced gas exchange and hypercapnia. The administered oxygen may also cause hypercapnia by suppressing the CNS respiratory drive. This is why oxygen therapy in patients with COPD should be done judiciously, with monitoring of the PaCO_2 .

Increased Pathologic Dead Space. A less common cause of hypercapnia with $\dot{V}/\dot{Q}$ mismatching occurs with increased pathologic dead space, where lung segments are ventilated but not perfused. What happens when pathologic dead space increases? Decreased effective ventilation will decrease gas exchange, and the CO_2 starts to build up in the alveolus and the blood, causing hypercapnia.

The most common way in which pathologic dead space causes hypercapnia occurs in **emphysema**, one of the types of COPD. Here, perfusion is reduced by both external compression of the pulmonary capillaries by overinflated alveoli and by destruction of the capillaries as part of the disease process (Figure 3), resulting in high $\dot{V}/\dot{Q}$ ratios.

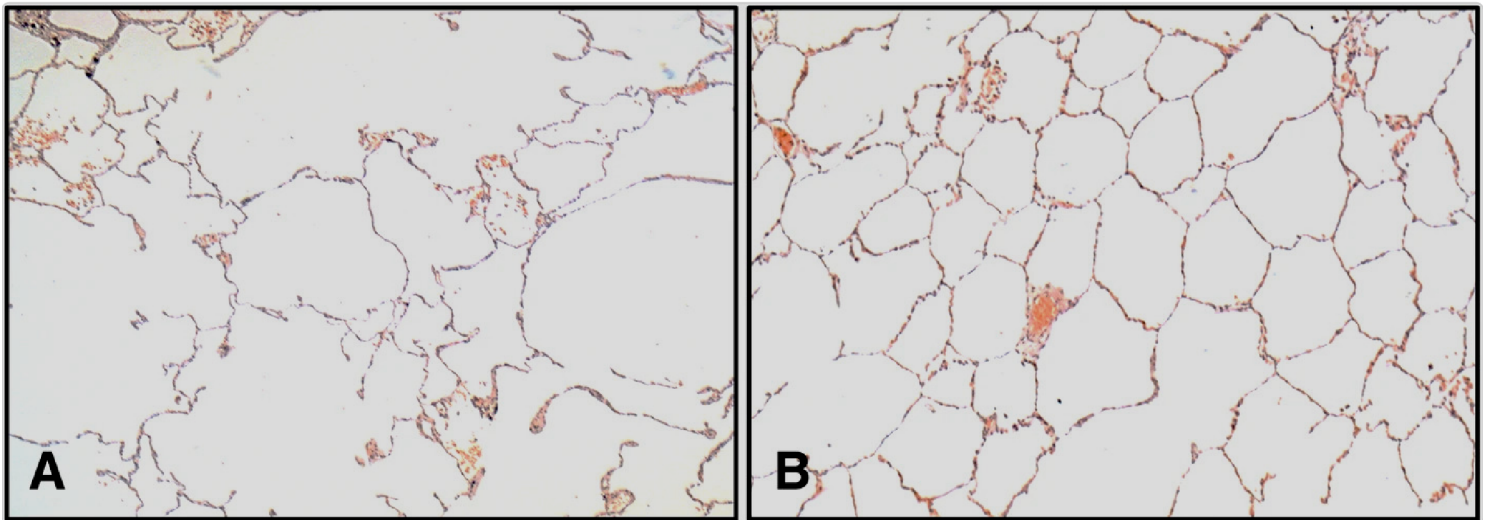


Figure 3 Destruction of alveolar walls and capillaries in Emphysema (A) vs normal lung (B)

Note that most people with COPD have a mixture of both bronchitis and emphysema. Thus, COPD can cause hypercapnia by reducing ventilation due to obstruction of the alveoli in bronchitis and by reducing perfusion because of capillary destruction in emphysema, making patients with COPD susceptible to hypercapnia.

Also note that a major cause of increased dead space, **pulmonary emboli (PE)**, **does not commonly cause hypercapnia**. The hypoxia caused by a PE stimulates a strong hyperventilation response, keeping the PaCO_2 normal or decreased in most cases.

decreased or compressed by enlarged alveoli, decreasing perfusion. Emphysema causes increased dead space because capillaries are

How Does the Body Respond to Hypercapnia (LO3)?

Metabolic (Renal) Response to Hypercapnia

The body compensates for hypercapnia with both hyperventilatory and metabolic responses. In hypercapnia, excess acid leads to **respiratory acidosis** because CO_2 reacts with water to form carbonic acid (H_2CO_3), which then dissociates into bicarbonate (HCO_3^-) and hydrogen (H^+) ions. As in any acid-base disturbance, the body compensates for the acidosis to try to keep the pH as normal as possible. In this case, the kidney provides the main compensation, retaining filtered bicarbonate to correct the low blood pH back toward normal.

Many of the symptomatic consequences of hypercapnia (e.g., cardiac conduction disturbances) are due to the acidemia, not to the high PaCO_2 per se. So, this renal compensation (especially prominent in patients with

chronic hypercapnia) is one reason why some patients with marked elevations of PaCO_2 may be asymptomatic.

Hyperventilatory Response to Hypercapnia

Increasing PaCO_2 can stimulate the central chemoreceptors in the brain's medulla as well as the peripheral receptors in the aortic arch and carotid arteries (Figure 1). Central chemoreceptors respond to H^+ directly, but since the blood–brain barrier blocks charged ions, it is CO_2 that diffuses across. Once inside, CO_2 forms carbonic acid, which then dissociates into H^+ (Figure 1).

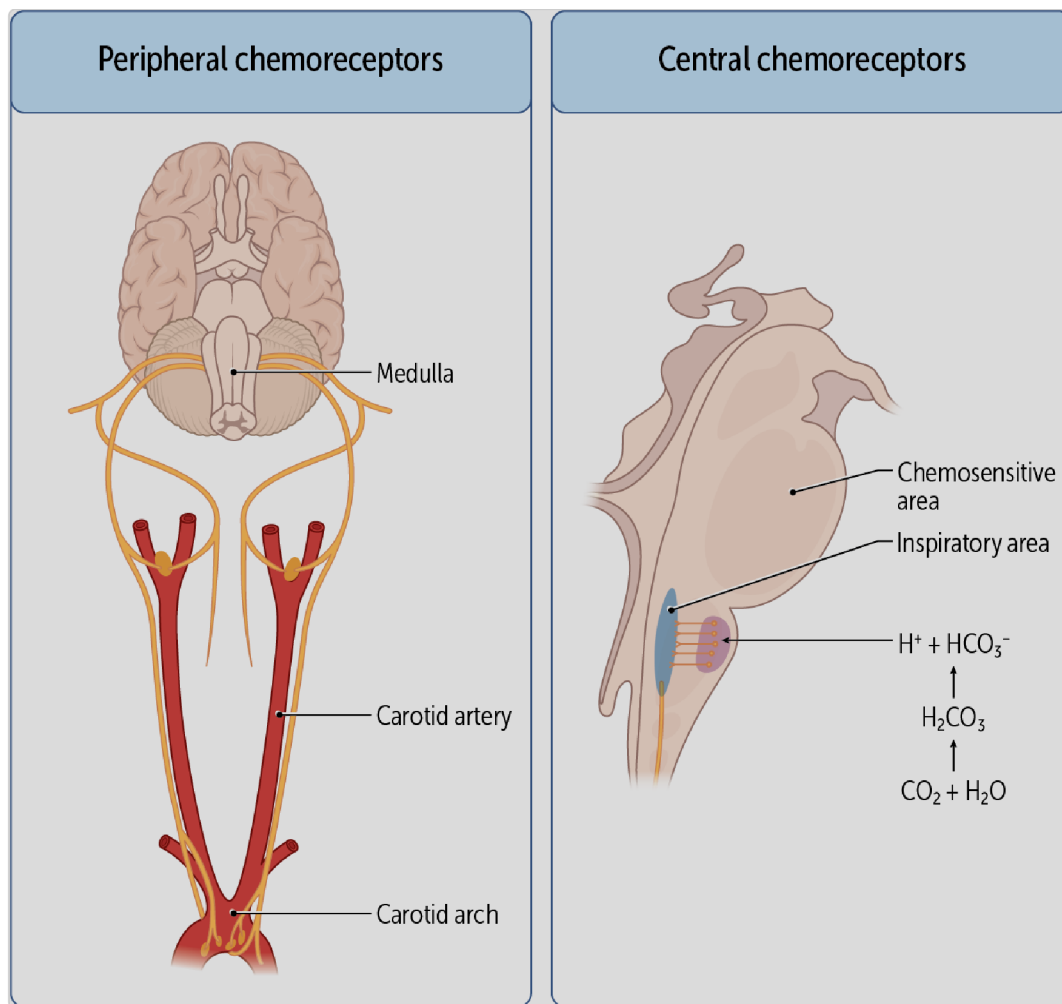


Figure 1 Peripheral (left) and central (right) chemoreceptors

These receptors then stimulate the respiratory center in the CNS and cause a higher respiratory rate and increased tidal volume. The result is **increased minute ventilation**, which drops the PaCO_2 back toward normal.

If the increased respiratory rate cannot keep up with the buildup of CO_2 , hypercapnia results. Why would this happen? Possibilities include the following:

- Over time, the body becomes habituated to the sensation of hypercapnia, such that it fails to stimulate the respiratory drive as much. This blunted response of the brain respiratory centers is more marked in patients

taking CNS depressants, like opioids. These patients may have asymptomatic, high values of PaCO_2 .

- The respiratory muscles can become fatigued, especially in patients who are chronically ill or malnourished, and this compensatory ventilation is less effective. This also occurs in rapidly breathing patients having an acute exacerbation of COPD or asthma. They can tire out due to the effort of breathing, leading to a rise in PaCO_2 .
- Patients with emphysema tend to take fast, shallow breaths. Because each breath must first fill the anatomic dead space, little air reaches the alveoli. This effect of the anatomic dead space makes these shallower breaths less efficient and reduces the expected rise in minute ventilation.

hypercapnia

emphysema can prevent the normal hyperventilatory response to respiratory muscle weakness, and shallow breathing seen in blunting of the brain respiratory center response (e.g. opiates),

What Are the Consequences of Hypercapnia (LO3)?

An increased PaCO_2 level has multiple consequences. In some cases, it's not clear whether the main culprit is the elevated CO_2 or the secondary drop in blood pH (respiratory acidosis), as they are linked biochemically. Both may contribute.

Central Nervous System Effects

Acute elevations in PaCO_2 will usually **depress a patient's level of consciousness** when it exceeds 75 to 80 mmHg. Severe acute hypercapnia can even lead to coma, cardiac or respiratory arrest, and death. It's a common reason to intubate patients in emergency departments.

In contrast, patients with chronic hypercapnia (eg, COPD) can usually tolerate higher PaCO_2 values without CNS changes. These patients will have elevated serum bicarbonate levels and more normal blood pH due to the renal compensation as discussed earlier. This suggests that the pH is at least partially responsible for the CNS effects. The effects of elevated CO_2 on the brain are not well understood but may involve altered levels of CNS neurotransmitters like glutamine and γ -aminobutyric acid (GABA).

Hypercapnia also causes vasodilation, with an increase in intracranial pressure. This is not usually severe enough to cause direct damage to the brain.

Cardiac Effects

In acute hypercapnia, arrhythmias and hypotension arise mainly from the accompanying acidosis rather than the elevated PaCO_2 itself. The excess H^+ reduces protein binding of calcium, increasing free ionized calcium. This

alters membrane excitability (a bathmotropic effect), impairing Na^+ entry through voltage-gated Na^+ channels and predisposing to cardiac dysrhythmias and vascular instability.

likely to cause CNS effects.

retention of bicarbonate with normalization of the pH, so it is less
Chronic hypercapnia leads to more compensations, including renal

How Do We Diagnose Hypercapnia (LO4)?

Because hypercapnia can be a fatal event or predict respiratory failure, effective detection of acute reversible causes is key. Hypercapnia should be looked for in any patient with respiratory complaints, especially if severe or progressive. It should also be evaluated for in patients with mental status changes of unknown cause.

Hypercapnia can be diagnosed using arterial or venous blood gas testing (the PaCO_2 and PACO_2 values are quite close) or capnography. **Capnography** uses a device attached to an endotracheal tube, nasal cannula, or face mask to measure inspired and expired CO_2 in real time. The End Tidal CO_2 (CO_2 level

in the expired gas at the end of expiration) is an excellent indirect estimate of the PACO_2 and therefore PaCO_2 . It can be displayed continuously without the need to draw blood, thus avoiding the discomfort of frequent arterial blood gasses. (Figure 4)

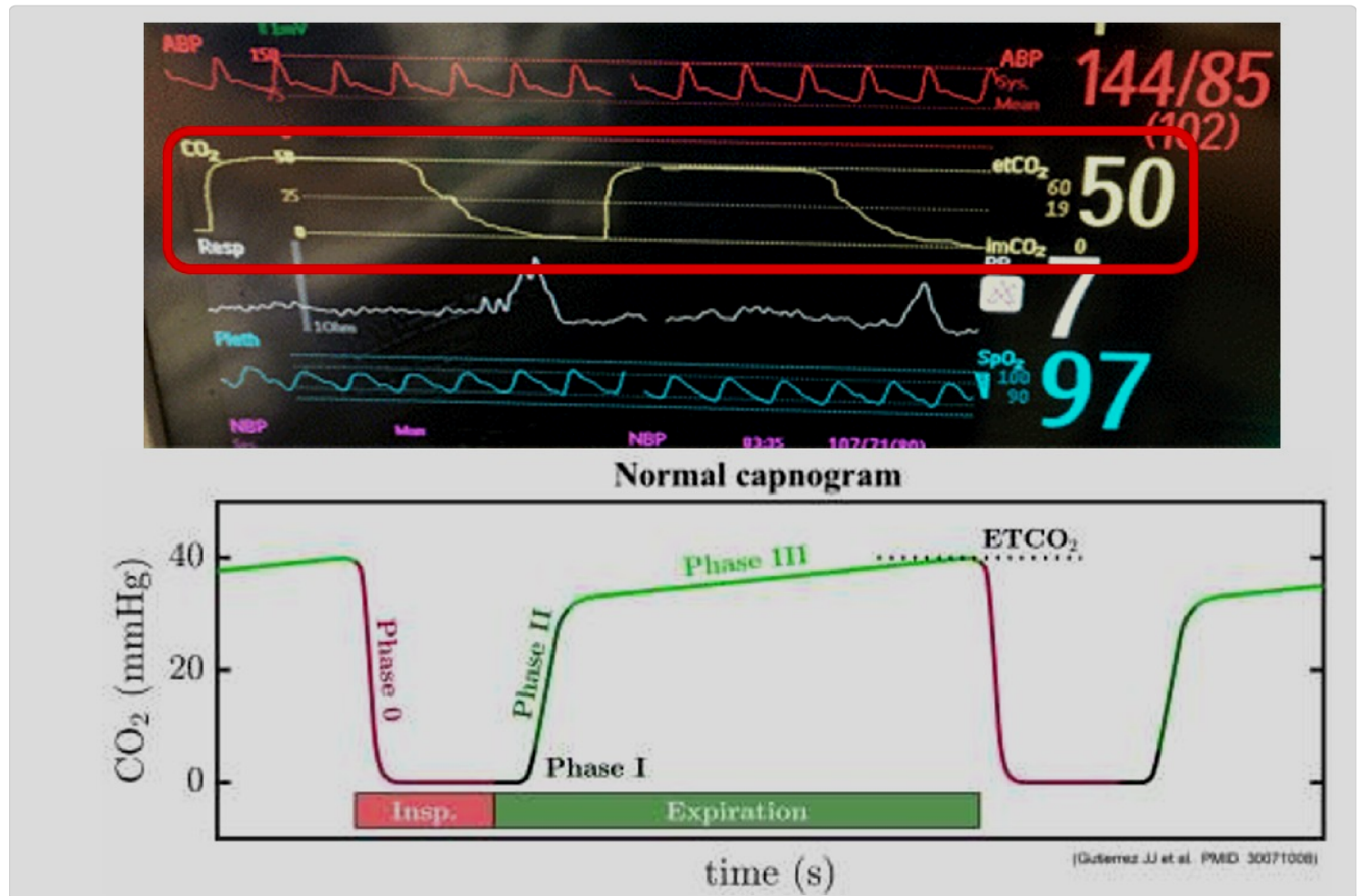


Figure 4 Capnography follows the End Tidal CO_2 (ETCO_2) as a continuous estimate of PaCO_2 .

pl00q.

Capnography monitors CO_2 in real time and does not require drawing

How Do We Manage Hypercapnia (LO5)?

Management of hypercapnia relies on first identifying and addressing the underlying cause to prevent future episodes or slow its progression in chronic patients.

Most critical is reversing acutely correctable causes, for example, in patients with hypercapnia due to substance overdose, an antidote like **naloxone** (opioid overdose) or **flumazenil** (benzodiazepines). All emergency department patients with an unknown cause of hypercapnia (and or decreased respiratory rate) should be given naloxone while the evaluation is initiated. Patients taking opioids should be given naloxone for home use and instructed on when to use it.

For patients with persistent hypercapnia (with or without hypoxia), mechanical ventilation is sometimes needed. We can start with noninvasive ventilation. Continuous positive airway pressure (CPAP- used for sleep apnea) and bilevel positive airway pressure (BiPAP) devices are both used to increase tidal volumes.

These modalities are particularly useful in those with respiratory failure due to cardiogenic pulmonary edema (i.e., heart failure), which can sometimes lead to hypercapnia due to $\dot{V}/\dot{Q}$ mismatch. BiPAP is sometimes used for

severe exacerbations of COPD. (Extra info: BiPAP provides 2 pressure levels, higher on inhale, lower on exhale, which supports tidal volume and reduces muscle fatigue in patients struggling to ventilate to attain a normal CO_2). Here, the increased airway pressures lead to increased tidal volumes and minute ventilation, resulting in a consequent decrease in respiratory rate and, therefore, a decrease in the work of breathing.

Invasive ventilation (e.g., endotracheal intubation) is required in sicker hypercapnic patients. The PaCO_2 measurement is not used by itself in the decision to intubate but is used together with the PaO_2 and overall patient symptoms, signs, and observed course. Specific numerical criteria for when to intubate do not exist—this is a judgment call requiring integration of all available data.

CLINICAL CORRELATION

While specific numerical criteria for intubation do not exist, certain trends strongly support the need for invasive ventilation in acute hypercarbia: severe acidosis (eg, $\text{pH} < 7.25$), worsening elevation of CO_2 , and respiratory muscle fatigue. Decreased consciousness may need intubation when it reaches a level of concern for depressed airway protective reflexes (gag, cough, avoiding aspiration), maintenance of a patent airway, and adequate ventilation.

benzodiazepines, respectively.
Naloxone and flumazenil are antidotes to opioids and

CASE CONNECTION

[BACK TO INTRODUCTION](#) 

Thinking back to NK, what explains his changed clinical condition after treatment?

While oxygen and bronchodilators were appropriate to treat his dyspnea and hypoxemia, oxygen carries a risk in patients with COPD. It can suppress the central drive to hyperventilate as well as cause increasing $\dot{V}/\dot{Q}$ mismatch by reversing the usual hypoxic vasoconstriction that shunts blood away from poorly ventilated areas of the lung. Together, this can lead to reduced air exchange and a rising PaCO_2 , even with an improvement in the PaO_2 . To avoid complications of hypercapnia, the emergency department staff reduced the administered oxygen to keep adequate oxygen saturation without causing PaCO_2 to rise. After 3 hours of treatment, NK was back to baseline and ready to go home.

Summary

- Hypercapnia is an elevation of the partial pressure of carbon dioxide (PaCO_2).

- It can be a sign of pulmonary, central nervous system (CNS), or musculoskeletal disease or of some overdoses.
- Patients with acute hypercapnia may be short of breath or have flushing or confusion.
- Patients with chronic hypercapnia may be asymptomatic.
- Elevated PaCO_2 is sensed by chemoreceptors, which stimulate increased minute ventilation (increased respiratory rate and tidal volume).
- The fall in pH (respiratory acidosis) is compensated by renal retention of bicarbonate.
- Decreased minute volume can be due to decreased CNS drive (e.g., opioids, CNS disease) or to respiratory muscle weakness with decreased tidal volume.
- Ventilation/perfusion ($\dot{V}/\dot{Q}$) mismatch due to fluid or mucus in the alveoli is seen in pulmonary disease and can lead to poor gas exchange and hypercapnia.
- Emphysema can decrease lung perfusion and gas exchange by increasing physiologic dead space from capillary destruction or compression.
- Hypercapnia can cause confusion, asterixis, or coma.
- Hypercapnia and the resultant respiratory acidosis can cause cardiac arrhythmias or hypotension if severe.
- Hypercapnia is diagnosed using blood gas or capnography measurements.
- Capnography measures CO_2 in real time and does not require a blood draw.
- Noninvasive or invasive ventilation can correct hypercapnia.
- Opioids or sedatives should be suspected in patients with acute hypercapnia and treated with antidotes promptly.

Review Questions

1. A 25-year-old male is wheeled into the emergency department because of a sudden loss of consciousness after a narcotic overdose. He has a temperature of 36°C, respiratory rate of 6 breaths/min with adequate depth on both lungs, heart rate of 120 beats/min, and blood pressure of 130/89 mmHg. Arterial blood gas results are pH 7.16; partial pressure of arterial oxygen, 59 mmHg; and partial pressure of arterial carbon dioxide (PaCO_2), 70 mmHg. What is the most likely cause of his high PaCO_2 ?

- A. Airway obstruction
- B. Decreased dead space
- C. Decreased minute ventilation
- D. Increased CO_2 production
- E. Increased dead space

Explanation (requires correct answer)

2. Which of the following best explains the development of chronic hypercapnia in patients with chronic obstructive pulmonary disease, a condition that typically involves elements of both chronic bronchitis and emphysema?

- A. Airway obstruction
- B. Decreased physiologic dead space
- C. Increased CO₂ production
- D. Ventilation/perfusion mismatch
- E. Increased minute ventilation

Explanation (requires correct answer)

3. Which of the following is least likely seen in a patient with hypercapnia?

- A. Asterixis
- B. Confusion
- C. Decreased blood pH
- D. Flushing
- E. Increased renal bicarbonate excretion

Explanation (requires correct answer)

- A. Chest X-ray

- B. Serum bicarbonate
- C. Capnography
- D. Pulse oximetry

Explanation (requires correct answer)

5. A 65-year-old man with COPD presents with altered mental status. Arterial blood gas reveals pH 7.28, PaCO₂ 72 mmHg, PaO₂ 65 mmHg. Which intervention is most appropriate to initiate first?

- A. Endotracheal intubation
- B. Increase supplemental oxygen to 100%
- C. Begin noninvasive ventilation with BiPAP
- D. Order a chest X-ray

Explanation (requires correct answer)

How would you rate this Brick?



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 Describe the clinical presentation of hypercapnia.

02 Explain the mechanisms underlying hypercapnia.

- 03** Explain the consequences of hypercapnia and the body's compensatory mechanisms.
- 04** Interpret clinical findings to properly diagnose hypercapnia.
- 05** Outline the management of hypercapnia.

Objective Confidence: 0/5